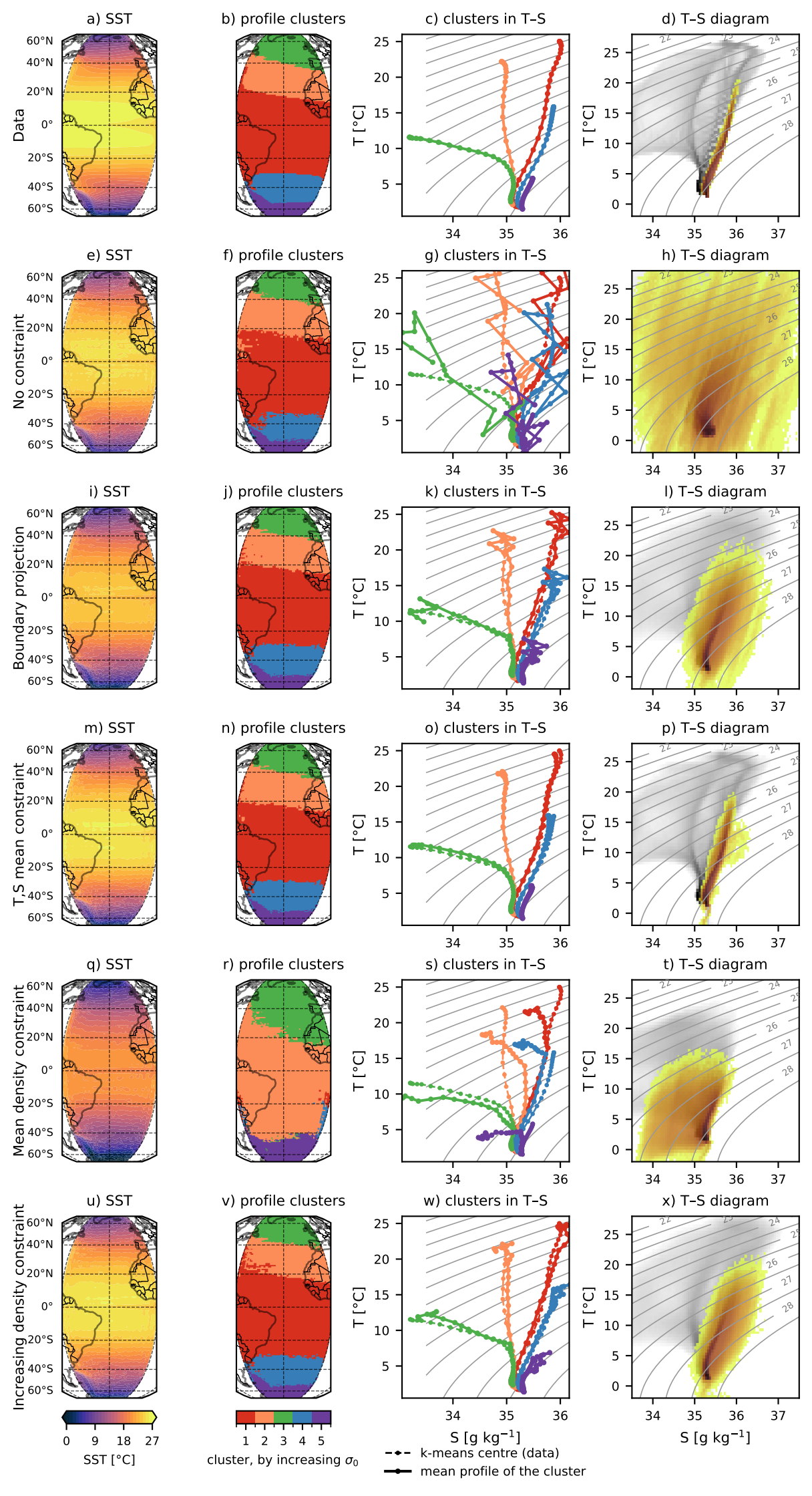

a) SST
b) profile clusters
c) clusters in T–S
d) T–S diagram
Data
e) SST
f) profile clusters
g) clusters in T–S
h) T–S diagram
No constraint
i) SST
j) profile clusters
k) clusters in T–S
l) T–S diagram
Boundary projection
m) SST
n) profile clusters
o) clusters in T–S
p) T–S diagram
T,S mean constraint
q) SST
r) profile clusters
s) clusters in T–S
t) T–S diagram
Mean density constraint
u) SST
v) profile clusters
w) clusters in T–S
x) T–S diagram
Increasing density constraint
60°N
40°N
20°N
0°
20°S
40°S
60°S
T [°C]
S [g kg$^{-1}$]
0 9 18 27
SST [°C]
1 2 3 4 5
cluster, by increasing $\sigma_0$
k-means centre (data)
mean profile of the cluster